Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Klemens Ilse	Email	klemens.ilse@csp.fraunhofer.de
Institute Name	Fraunhofer Institute for Microstructure of Materials and Systems IMWS			
Address	Walter-Huelse-Str 1, D-06120 Halle, Germany			

2. Technology Overview

Title of Proposed Technology	Optical quality inspection, anomaly detection, and pick-and-place orientation for advanced manu- and valuefacturing			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Knowledge Services		Mechanics Materials	
	Medium Category	Small Category	Medium Category	Small Category
	R&D / Engineering Services	Testing / Inspection / Analysis Techniques	Robotics/Automation Machinery	Robot Vision and Production Automation Technologies
Concept of Proposed Technology	AI-based camera classification to automate end-of-line and in-process optical inspection across large product variety and small defects			
Rationale for Joint R&D Project	Korean SMEs gain an industrial-grade, co-developed and validated in real production environments. Joint R&D reduces manual inspection effort, increases quality and throughput, and transfers know-how into robust, replicable modules tailored to local lines			

(max 10 pages)

Key Words	Optical quality inspection Anomaly detection and Instance segmentation AI-based image classification and advanced manufacturing

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	-Deliver pilot-ready, AI-based camera module for optical inspection, anomaly detection, segmentation, and for example pick and place orientation. - Demonstrate reliable performance across large product variety and very small defects that may be visible at pixel level. - Shorten time to value through synthetic data generation and one-click annotation, starting from small image sets. - Provide a transferable package including datasets, trained models, operating procedures, and success measures.			
R&D Target Specifications	Functional: image classification (training data < 15 images), object and defect detection, anomaly detection, semantic and instance segmentation, text reading through optical character recognition –> 95% faster annotation Operation: fully offline deployment through the application programming interface; automatic handling of large images; region-of-interest processing; enterprise features such as role-based annotations and training with weighted annotations.			
R&D Plan	WP1 – Use-case definition: define product scope, defect types, acceptance criteria, quality and throughput targets, and labor reduction targets. WP2 – imaging and data preparation: design imaging workflow (for example, monochrome, color, or hyperspectral imaging), collect data, and set up data quality checks; prepare ground truth with the one-click annotation tool and region-of-interest guidance. WP3 – Synthetic data and augmentation: generate AI images and apply meaningful transformations to cover product variety and rare defects. WP4 – Model development: perftrain models for classification, detection, anomaly detection, segmentation, text reading, and matching; f.e. enable rotating bounding boxes with orientation for pick and place. WP5 – Deployment and integration: package modules for fully offline use programming interface; integrate into the production environment; prepare monitoring and quality checks. WP6 – Pilot and validation: run the pilot in a production-like setting; evaluate accuracy, robustness to product variety, and ability to detect very small defects; document results and train operators.			

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	○ methodology and pipeline design: architecture search, ROI/instance segmentation, anomaly detection, OCR/code reading, matching ○ Data acquisition concept, annotation guidelines, quality gates, and KPIs/metrics definition; validation and test protocols. ○ Pilot implementation and evaluation; documentation, trainings, and technology transfer packages ○ use-case spec & KPIs, curated/augmented datasets, annotated ground truth, trained models
Expected R&R of Korean SME Partner	○ Provide production samples/data and domain expertise; define acceptance criteria and operational constraints (takt time, interfaces) ○ Provide production samples/data and domain expertise; define acceptance criteria and operational constraints (takt time, interfaces) ○ pilot environment and support integration to existing equipment/IT; provide feedback loops and operators for evaluation ○ Provide resources for data collection/annotation; support maintenance during the pilot

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	- Production-grade pilot at Level 6 that runs fully offline and improves quality and production output while reducing manual end-of-line testing; - Reliable handling of large product variety and very small defects; strong performance for text reading, code reading, matching, and pick and place orientation. - Short ramp-up due to synthetic data generation and one-click annotation; reusable blueprint for further products - Transferable outcomes including datasets, trained models, operating procedures, and validation reports that enable scaling.

6. References

(1)

(2)

(max 10 pages)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

(max 10 pages)